

In-body Antigen Detection via Antibodies on a Transistor

A Simulation Study of HER2 Transport, GFET Response, and Feasibility Limits

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Main finding

Directional flow and sensor placement improve HER2 exposure, but electrostatic screening and electronics noise dominate practical GFET detectability.

Problem

HER2 is an important breast cancer biomarker, but continuous in-body monitoring is not established. This project evaluates whether a GFET biosensor concept could detect HER2 after transport, binding, and electrical transduction are considered together.

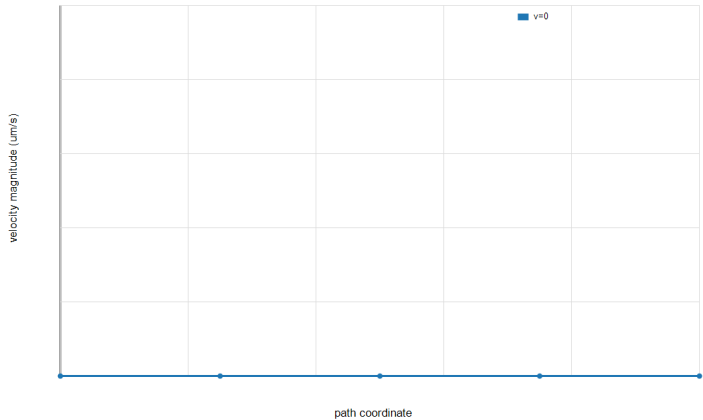
Method Pipeline

- HER2 concentration input
- Diffusion or prescribed directional flow
- Local sensor exposure
- Antibody binding and bound molecule count
- GFET current response
- LOD, detectability, and Debye screening tests

COMSOL Model

- M2: diffusion-only Transport of Diluted Species baseline.
- M2B: partially anatomical prescribed-velocity convection-diffusion extension.
- Verification: $v_{in} = 0$ behaves like diffusion-only; increasing v_{in} changes exposure.
- Limitation: M2B is not a full anatomical lymph-node model and not a validated Darcy-flow pressure solution.

M2B prescribed velocity streamlines

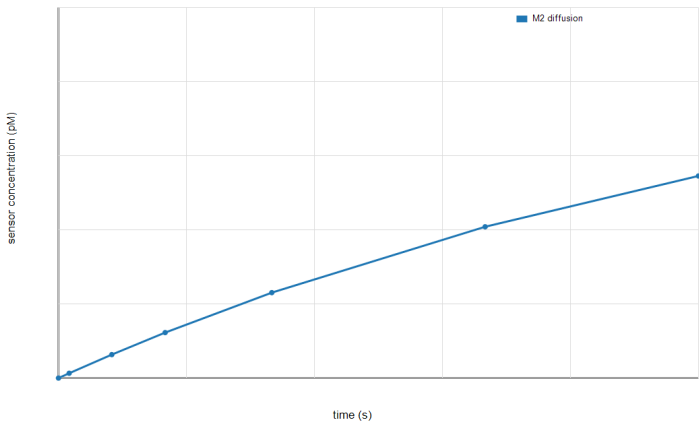


Prescribed M2B velocity field.

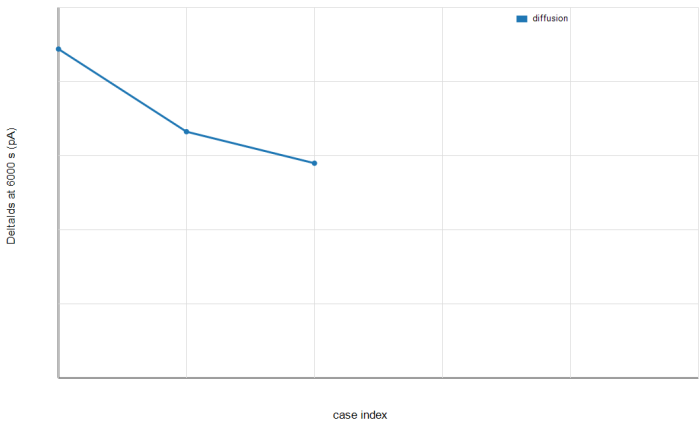
Result 1: Flow and Placement

- At $C = 10$ pM, reference flow $v_{in} = 5 \times 10^{-7}$ m/s gives 8.73 pM at 6000 s.
- Diffusion-only transport gives 4.98 pM at 6000 s.
- Time-to-50% exposure decreases from about 6038 s to 1839 s.
- Subcapsular/cortical placement is fastest and strongest under reference flow.

Experiment A: flow vs diffusion exposure



Experiment B: placement current response

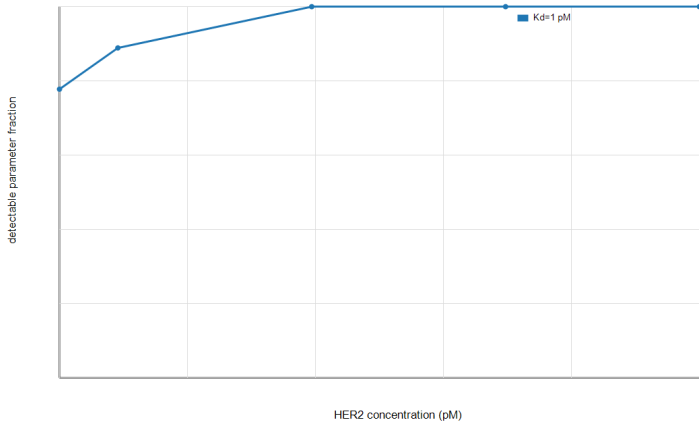


Result 2: Current Response

- GFET current response increases with HER2 concentration, coupling efficiency, and bound molecule count.

- Experiment C finds 99 detectable rows and 36 failing rows.
- Best reported M4 LOD is approximately 0.78 pM under $\alpha = 0.03$ and 10 pA noise.
- Sub-pM detectability is conditional, not universal.

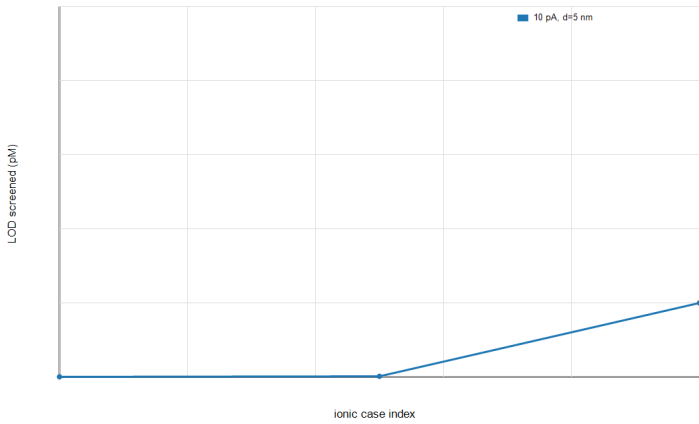
Experiment C: detectable fraction



Result 3: Debye Screening

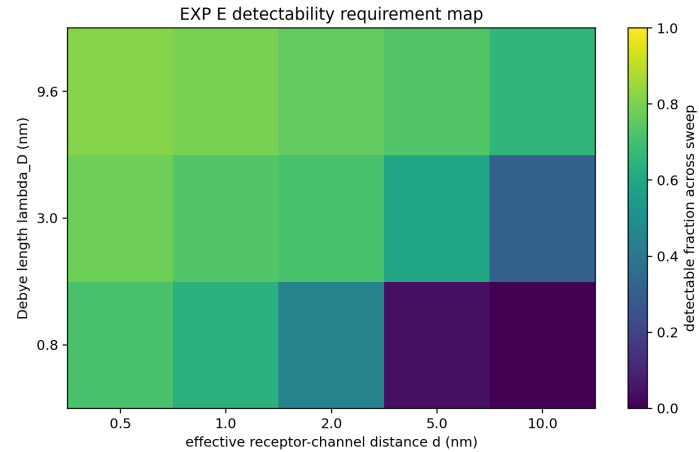
- Screening uses $\alpha_{eff} = \alpha_0 \exp(-d/\lambda_D)$.
- PBS-like physiological salt with $\lambda_D = 0.8$ nm and $d \geq 5$ nm gives 0/60 detectable screened cases.
- Screening and electronics noise are the main feasibility barriers.

Experiment D: LOD vs ionic strength



Result 4: Design Requirements

- Feasibility requires shorter receptor-channel distance, stronger coupling, lower current noise, or reduced ionic screening.
- At $K_d = 10$ pM and 10 pA noise under PBS-like screening at $d = 1$ nm, required α_0 is about 0.0326 at 1 pM HER2 and 0.00561 at 10 pM HER2.
- Experiment E turns the Debye-screening limitation into quantitative requirements.



Limitations

- Simulation-only; no wet-lab or clinical validation.
- No fabricated device is claimed.
- Flow is prescribed, not a validated pressure solution.
- GFET response is analytical rather than a full graphene device simulation.

Future Work

- Experimentally validate these design requirements.
- Replace prescribed flow with validated Darcy/Brinkman flow.
- Automate direct COMSOL field exports.
- Test surface chemistries/electronics that can achieve the required distance, coupling, and noise values.